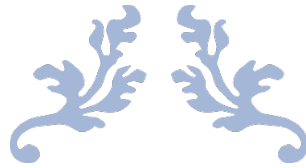


Nour "Eldeen" Ali

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DATA STRUCTURE

Final Project



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Team Members:

1. Nour Eldeen Ali Mohamed Ali Hassan El-Khouly
نور الدين علي محمد علي حسن الخولي
Group 4 “General”
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2. Marwan Ehab Adel Mostafa
مروان ايهاب عادل مصطفى
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مروان احمد السيد المهدي
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4. Mo’men Osama Galal Awad
مؤمن اسامة جلال عوض
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5. Youssef Ramadan Ahmed Mohamed El-Gamal
يوسف رمضان احمد محمد الجمل
Group 4 “General”
ID: 2401247684

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An Example of our
Code: we have created
a dummy data and
tried to list the courses
using the 9th function

```
9-List Courses By Students
10-List Students By Courses
11-Sort Students By ID
12-Sort Courses By ID
13-Is Full Course ?
14-Is Normal Student ?
15-Undo Last Action
16-Redo Last Action
0-Exit
```

```
Enter your choice:
```

```
9
```

```
Enter Student's ID :
```

```
1
```

```
Courses that Nour has enrolled in :
```

```
Calculus
```

```
Linear Algebra
```

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Roles:

1. Data Structures Creation:

Nour “Eldeen” Ali

Youssef Ramadan El-Gamal

Created the Students Nodes and Lists and Some of their functions

2. Functions:

Mo'men Osama Galal

Marwan Ahmed El-Mahdi

Marwan Mohamed

Created the main functions which were used in the ui of the console

3. Undo/Redo Stacks:

Marwan Ehab Adel

Implemented the Undo/Redo functionalities from scratch, from building their stacks to making them work and change in the structure of the sparse table

4. Console Menu:

*The amazing console menu was made by **Marwan Ahmed El-Mahdi**, giving a clear view of all the options*